N O N - N O R M A T I V E

T = sigma_v det = -1 (x, y) -> (-x, y)

"This reflects the standard."

Informative / Advisory (NOTE)

Owl Semaphore — Non-Normative

OWL 2 / Reflection State (sigma_v) / Standard Specification

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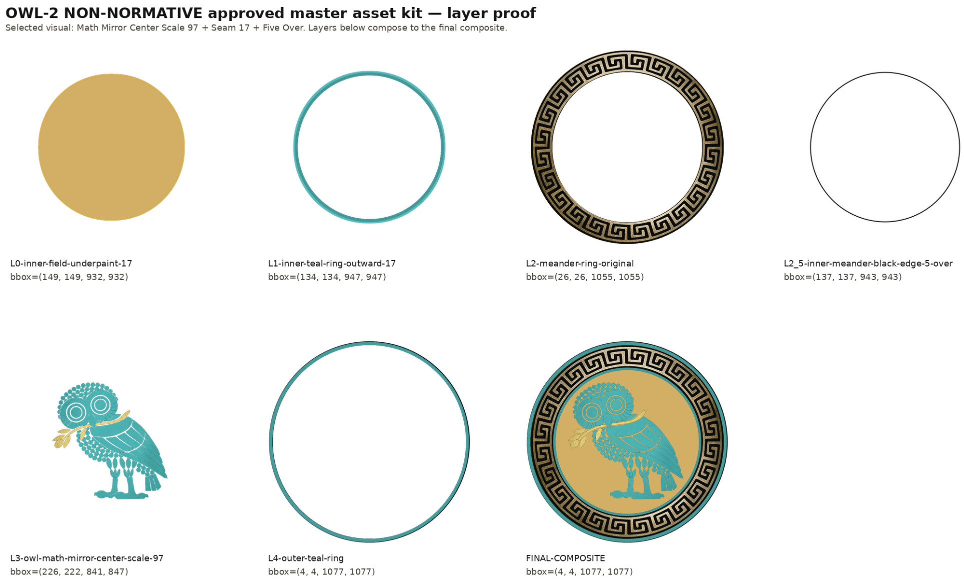
SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v2.0.1 · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=NON-NORMATIVE :: TRANSFORM=T = sigma_v det = -1 (x, y) -> (-x, y) :: QUOTE="This reflects the standard." :: VERSION=v2.0.1 :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PUBLISHED-VERSION-DOI=10.5281/zenodo.20418539 :: VERSION-DOI=TBD_BY_ZENODO_ON_V2_0_1_RELEASE

NON-NORMATIVE — LAYER PROOF PALETTE



1. Statement of Intent

This document defines the **NON-NORMATIVE owl** as a formal epistemic state within the Owl Semaphore system.

This is not a decorative variation of the normative mark. It is a mathematically defined reflection state with a specific role in analysis, interpretation, and structured disagreement.

The goal is to preserve rigor while allowing controlled deviation from the baseline framework.

1A. The Story Before the Math — *Da Vinci's Wings*

$$\mathbf{T = \sigma_v \quad \cdot \quad det = -1 \quad \cdot \quad (x, y) \rightarrow (-x, y)}$$

Stand in front of a mirror. You are still upright — up is still up, down is still down — but left and right have swapped. That is a vertical-axis reflection: $(x, y) \rightarrow (-x, y)$. Not a rotation — a mirror image.

When Leonardo da Vinci spent years studying birds and sketching wing mechanics — preserved in the *Codex on the Flight of Birds* and related manuscripts now held by the Biblioteca Reale di Torino (Library of Congress, *Codex on the Flight of Birds*; Smithsonian Air and Space, "Leonardo da Vinci's Codex on the Flight of Birds") — he was facing the other direction from the prevailing

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assumption that powered human flight was beyond reach. His machines did not become practical aircraft. What he left behind was a rigorous exploratory record: anatomical observation, geometric reasoning, and structured speculation.

The first powered, sustained, controlled heavier-than-air flight is attributed to the Wright Brothers at Kitty Hawk on 17 December 1903 (Library of Congress, Wright Brothers collection; Smithsonian National Air and Space Museum, *1903 Wright Flyer*). That success arrived four centuries after Leonardo, inside an accumulated aeronautical history of gliders, propulsion experiments, and aerodynamic theory — not as a direct line from his notebooks, but inside the broader practice of non-normative work that had matured into something actionable. The historical claim made here is the modest one: structured, rigorous exploration that has not yet succeeded is still part of the engine of progress.

The NON-NORMATIVE owl marks that posture — the mirror state, facing what the canonical view has its back to, **without claiming to have replaced the canonical view yet.** It is reflection, not rejection. Non-normative work is the engine of progress: rigorous exploration that has not finished yet.

This story is the human-intuition bridge to the mathematical formalism in §§2 onward. The deliberate ordering is story → transform → scientific use → objections/verification, so a reader who would argue the σ_v state from intuition, from history-of-science, or from the V_4 algebra can each enter through the right door.

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2. System Context

The Owl Semaphore system is defined by the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

The NON-NORMATIVE owl corresponds to the vertical reflection operator:

$$\sigma_v : (x, y) \mapsto (-x, y)$$

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3. Ontological Role

3.1 Semantic Designation

NON-NORMATIVE represents:

- legitimate alternative interpretation
- reflective critique
- analytical deviation
- structured disagreement

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3.2 Interpretive Role

This state indicates that the content:

- departs from the canonical model
- remains structurally grounded in the same system
- is not arbitrary or invalid

3.3 What It Does Not Mean

- not random
- not incorrect by default
- not adversarial (that is CRITICAL)

It is **reflection**, not rejection.

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4. Mathematical Definition

4.1 State Operator

$$T_{\text{non}} = \sigma_v$$

4.2 Matrix Form

$$\sigma_v = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

4.3 Determinant

$$\det(\sigma_v) = -1$$

4.4 Properties

- orientation-reversing
- reflection class
- order 2

$$\sigma_v^2 = I$$

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5. Coordinate System

The NON-NORMATIVE state uses the same coordinate system as NORMATIVE:

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- canvas: 1080 × 1080
- center: (540, 540)

Transformation is applied relative to this center.

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6. Canonical Orientation

6.1 Visual Definition

- upright
- faces LEFT

6.2 Transform Relationship

The NON-NORMATIVE owl is the horizontal mirror of the normative owl.

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7. Asset Topology

Layer structure is identical to normative:

- L1 — inner field
- L2 — meander ring
- L3 — owl body
- L4 — outer ring

7.1 Composite Definition

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

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8. Geometry

All geometric constraints are inherited from the normative standard:

- identical radii
- identical center
- identical annular structure

No geometric deformation is permitted.

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9. Color Specification

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9.1 Palette

- outer ring: #316964 (teal)
- owl: #316964 (teal)
- field: #d2d8d6 (cool gray)
- meander: unchanged (gold)

9.2 Color Doctrine

Teal represents analytical distance from the normative baseline while maintaining structural coherence.

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10. Transparency and Alpha

Same rules as normative:

- RGBA required
- corner alpha = 0
- center alpha = 255

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11. Provenance

11.1 Construction

Derived from normative by:

- horizontal reflection
- no rotation
- no scaling

11.2 Transform Integrity

The transform must be exact. Any distortion invalidates the state.

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12. Asset Invariants

12.1 Algebraic

- operator = σ_v
- determinant = -1

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12.2 Visual

- upright
- mirrored orientation

12.3 Structural

- geometry unchanged
- layer structure unchanged

13. Integrity Regime

All assets must:

- pass SHA-3-512 verification
- be reproducible from layers

14. Interpretation Rules

14.1 Positive Rule

When present:

The content reflects the normative framework while offering a structured alternative.

14.2 Negative Rule

It does not indicate failure or error.

15. Non-Permitted Changes

- rotation
- vertical flip
- C2 inversion
- geometry alteration

16. Relationship to Other States

- NORMATIVE → identity (*"This is the standard."*)
- NON-NORMATIVE → reflection (*"This reflects the standard."*)

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- CRITICAL → inversion (*"This inverts the standard."*)
- METACOGNITIVE → frame-audit (*"The observer audits the frame."* — thinking examines its own frame)

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17. Formal Definition

Let L_1–L_4 be the layer fields.

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

with

$$T = \sigma_v$$

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18. Closing Statement

The NON-NORMATIVE owl preserves system integrity while enabling structured deviation.

It is the mechanism by which the system permits disagreement without collapse.

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OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v2.0.1)

NORMATIVE

$T = I \quad \det = +1$

(x, y) -> (x, y)

"This is the standard."

RFC 2119 MUST / SHALL

NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

(x, y) -> (-x, y)

"This reflects the standard."

Informative / Advisory (NOTE)

CRITICAL

$T = C_2 \quad \det = +1$

(x, y) -> (-x, -y)

"This inverts the standard."

RFC 2119 MUST NOT / SHALL NOT

METACOGNITIVE

$T = \sigma_h \quad \det = -1$

(x, y) -> (x, -y)

"The observer audits the frame."

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).

Owl Semaphore v2.0.1 · github.com/IT-Help-San-Diego/owl-semaphore

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